Network’s Lab.

Instructor: Masoud Asghari

Session: 4

Save Switches config

- After any config to switches and routers
- In case of "power cycle devices"
  - You will see the change does not persist

- To keep config changes:
  - You need to write running config to startup config by:
  - #write
  - Or
  - #copy running-config startup-config

Show command in *En* mode

- #show ?
- #show clock
- #show interfaces
- # Show Interface [interface name]
  - # Show Interface [interface name]
  - #show interfaces fastEthernet 0/1
  - #show interfaces vlan 1
- #show ip interface brief
- #show version
- #show startup-config
- #show running-config
  - Copy all lines from putty to a file
  - You can paste it to putty at (Config)#

Config interfaces in a switch

- #conf t
- (config)#interface ?
- (config)#interface gigabitEthernet 0/1
  - (config)#interface FastEthernet 0/1
- (config-if)#shutdown
- (config-if)#no shutdown
- (config-if)#speed 10
- (config-if)#speed auto
- (config-if)#switchport ?
- (config-if)#exit

VLAN introduction

- A virtual LAN (VLAN)
  - is any **broadcast domain** that is partitioned and isolated in a computer network at the **data link layer** (OSI layer 2)
- VLANs Allow
  - **Group hosts together** even if the hosts are not on the same network switch.
  - issues such as scalability, security, and network management.
  - VLANs filter broadcast traffic, enhance network security, perform address summarization, and mitigate network congestion.
- To subdivide a network into virtual LANs,
  - Simpler network devices can only partition per physical port
  - More sophisticated devices can mark packets through tagging,
    - A single interconnect (**trunk**) may be used to transport data for multiple VLANs.
- Any switch port can belong to a VLAN, and unicast, broadcast, and multicast packets are forwarded and flooded only to end stations in that VLAN.
- Each VLAN is considered as a **logical network**, and packets destined for stations that do not belong to the VLAN **must be forwarded through a router.**